QUASI-HAMILTONIAN STRUCTURES ON EQUIVARIANT SURFACE REPRESENTATION SPACES AND THEIR TWISTED FIXED-POINT SPACES

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Allowed standard prerequisites (editorial).

- Alekseev-Malkin-Meinrenken quasi-Hamiltonian/loop-group correspondence. AMM, author cites Theorems 8.3 and 9.1 with source-local construction retained
- Sobolev gauge theory on compact surfaces with boundary. Author section 5 and bibliography

1. Finite group actions on quasi-Hamiltonian spaces

In this section, we give the definition of a finite group Γ-compatible q-Hamiltonian G-space and present a basic example that is an analogue of the double.

1.1. **Compatibilities of bitorsors.** Let G be a compact Lie group with Lie algebra $\mathfrak{g}$ and let Γ be a finite group. We fix an action $\Gamma \to \mathrm{Aut}(G)$ of Γ on G. Also we give a G-bitorsor $\mathbb{G}$ equipped with a left Γ-action where a bitorsor $\mathbb{G}$ is a manifold endowed with left and right G-actions that are both simply transitive and commute with each other.

Definition 1.1. A Γ-compatible G-bitorsor $\mathbb{G}$ is a bitorsor $\mathbb{G}$ equipped with a Γ-action such that $\varphi \cdot (gx) = (\varphi \cdot g)(\varphi \cdot x)$, $\varphi \cdot (xg) = (\varphi \cdot x)(\varphi \cdot g)$, ($\varphi \in \Gamma$, $g \in G$, $x \in \mathbb{G}$).

Example 1.2. We define the space $\mathbb{G} := \mathrm{Map}(\mathbb{Z}/m\mathbb{Z}, G)$. Fixing an action $\varphi\colon \mathbb{Z}/m\mathbb{Z} \to \mathrm{Aut}(G)$, we define a $\mathbb{Z}/m\mathbb{Z}$-action on $\mathbb{G}$ by $(l \cdot g)(k) := \varphi(l)g(k-l)$, $l, k \in \mathbb{Z}/m\mathbb{Z}$, $g \in \mathbb{G}$. Then the space $\mathbb{G}$ is a $\mathbb{Z}/m\mathbb{Z}$-compatible $\mathrm{Map}(\mathbb{Z}/m\mathbb{Z}, G)$-bitorsor.

Remark 1.3. Let $\mathbb{G}$ be a Γ-compatible G-bitorsor. Define left and right actions of $G \rtimes \Gamma$ on $\mathbb{G} \times \Gamma$ by $(g, \varphi)(x, \psi) := (g(\varphi \cdot x), \varphi\psi)$, $(x, \psi)(g, \varphi) := (x(\psi \cdot g), \psi\varphi)$, where $(g, \varphi) \in G \rtimes \Gamma$ and $(x, \psi) \in \mathbb{G} \times \Gamma$. Then these actions make $\mathbb{G} \times \Gamma$ into a $G \rtimes \Gamma$-bitorsor. We denote this bitorsor by $\mathbb{G} \rtimes \Gamma$.

We define the inverse $\mathbb{G}^{-1}$ of a G-bitorsor $\mathbb{G}$ by the space $\mathbb{G}^{-1} := \left\{x^{-1} \mid x \in \mathbb{G}\right\}$, which is a copy of $\mathbb{G}$, equipped with the following G-actions, $gx^{-1} := \left(xg^{-1}\right)^{-1}$, $x^{-1}g := \left(g^{-1}x\right)^{-1}$ for $g \in G$ and $x \in \mathbb{G}$.

Moreover, we define the product $\mathbb{G}_1 \cdot \mathbb{G}_2$ of two G-bitorsors $\mathbb{G}_1$ and $\mathbb{G}_2$ as the quotient of $\mathbb{G}_1 \times \mathbb{G}_2$ by the following action, $g(x_1, x_2) := \left(x_1 g^{-1}, g x_2\right)$ for $g \in G$ and $(x_1, x_2) \in \mathbb{G}_1 \times \mathbb{G}_2$. We denote an element $[x_1, x_2] \in \mathbb{G}_1 \cdot \mathbb{G}_2$ by $x_1 \cdot x_2$.

Proposition 1.4. *Let $\mathbb{G}$, $\mathbb{G}_1$, $\mathbb{G}_2$ be Γ-compatible G-bitorsors. Suppose that the fixed point sets $\mathbb{G}^\Gamma$, $\mathbb{G}_1^\Gamma$, $\mathbb{G}_2^\Gamma$ are non-empty. Then we have the following properties:*

1. *The fixed point set $\mathbb{G}^\Gamma$ becomes a G^Γ-bitorsor under the restriction of the action of G on $\mathbb{G}$.*
2. *The inverse $\mathbb{G}^{-1}$ of the bitorsor $\mathbb{G}$ and the product $\mathbb{G}_1 \cdot \mathbb{G}_2$ of $\mathbb{G}_1$ and $\mathbb{G}_2$ become Γ-compatible G-bitorsors.*

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(3) *There exists an isomorphism of G^Γ-bitorsor* $\mathbb{G}_1^\Gamma \cdot \mathbb{G}_2^\Gamma \simeq (\mathbb{G}_1 \cdot \mathbb{G}_2)^\Gamma$.

Proof. Statement (1) is easy to verify. We define actions of Γ on $\mathbb{G}_1^{-1}$ and $\mathbb{G}_1 \cdot \mathbb{G}_2$ by $\varphi x^{-1} := (\varphi x)^{-1}$, $\varphi(x_1 \cdot x_2) := (\varphi x_1) \cdot (\varphi x_2)$, for $\varphi \in \Gamma$ and $x_j \in \mathbb{G}_j$. These actions satisfy the definition of Γ-compatibility.

To show statement (3), we see that the canonical map $\Phi\colon \mathbb{G}_1^\Gamma \cdot \mathbb{G}_2^\Gamma \to (\mathbb{G}_1 \cdot \mathbb{G}_2)^\Gamma$, $x_1 \cdot x_2 \mapsto x_1 \cdot x_2$ is equivariant with respect to the left and right G^Γ-actions. For any $g \in G^\Gamma$, $x_1 \cdot x_2 \in \mathbb{G}_1^\Gamma \cdot \mathbb{G}_2^\Gamma$,

$$\Phi(g(x_1 \cdot x_2)) = \Phi(gx_1 \cdot x_2) = gx_1 \cdot x_2 = g(x_1 \cdot x_2) = g\Phi(x_1 \cdot x_2).$$

The equivariance of Φ with respect to the right G^Γ-action is verified in the same way as above. Hence, by the simple transitivity of the left and right actions on bitorsors, Φ is an isomorphism. □

1.2. **Compatibilities of q-Hamiltonian G-spaces.** First, we recall the definition of twisted quasi-Hamiltonian G-spaces [5, 15]. Throughout this paper, we fix an Ad-invariant positive definite inner product $(\ ,\ )$ on $\mathfrak{g}$ and consider only bitorsors whose corresponding elements in $\mathrm{Out}(G)$ preserve this inner product. Also we assume that the Γ-action on $\mathfrak{g}$ induced by the action on G preserves the inner product on $\mathfrak{g}$.

We define the left Maurer–Cartan form θ and the right Maurer–Cartan form $\overline{\theta}$ on a G-bitorsor $\mathbb{G}$ by $\theta_x(X) := x^{-1}X$, $\overline{\theta}_x(X) := Xx^{-1}$, $x \in \mathbb{G}$, $X \in T_x\mathbb{G}$, where $x^{-1}X \in \mathfrak{g}$ denotes the derivative of the map $x^{-1}\colon \mathbb{G} \to G$, $y \mapsto x^{-1}y$ for X.

Definition 1.5. A (twisted) quasi-Hamiltonian G-space (M, ω, μ) is a G-manifold M together with an invariant 2-form ω and a G-equivariant map $\mu\colon M \to \mathbb{G}$ to a G-bitorsor $\mathbb{G}$ such that

(QH1) $\mathrm{d}\omega = \mu^* \frac{1}{12}(\theta, [\theta, \theta])$.
(QH2) $\iota\big(\xi^\#\big)\omega = \frac{1}{2}\mu^*\big(\theta + \overline{\theta}, \xi\big)$, $(\xi \in \mathfrak{g})$, $\xi^\#|_p = \frac{\mathrm{d}}{\mathrm{d}t}\big(\mathrm{e}^{-t\xi} \cdot p\big)|_{t=0}$.
(QH3) $\ker \omega_p \cap \ker d\mu_p = \{0\}$, $(p \in M)$,

where θ (resp. $\overline{\theta}$) denotes the left (resp. right) Maurer–Cartan form on $\mathbb{G}$.

We note that the restriction of the inner product $(\ ,\ )$ to the Lie algebra $\mathfrak{g}^\Gamma$ of the Γ-fixed point Lie subgroup $G^\Gamma := \{g \in G \mid \varphi \cdot g = g \text{ for all } \varphi \in \Gamma\}$ is also positive definite and invariant. Now we give the definition of a Γ-compatible quasi-Hamiltonian G-space.

Definition 1.6. A Γ-compatible quasi-Hamiltonian G-space (M, ω, μ) is a twisted quasi-Hamiltonian G-space equipped with a Γ-action such that

(1) $\varphi \cdot (g \cdot p) = (\varphi \cdot g) \cdot (\varphi \cdot p)$, $(\varphi \in \Gamma,\ g \in G,\ p \in M)$.
(2) The q-Hamiltonian 2-form ω is Γ-invariant.
(3) The moment map $\mu\colon M \to \mathbb{G}$ is Γ-equivariant.

We say that any two Γ-compatible quasi-Hamiltonian G-spaces are isomorphic if there exists a Γ-equivariant isomorphism of twisted quasi-Hamiltonian G-spaces.

From a Γ-compatible quasi-Hamiltonian G-space, one can naturally construct a twisted quasi-Hamiltonian $G \rtimes \Gamma$-space.

Remark 1.7. Let (M, ω, μ) be a Γ-compatible quasi-Hamiltonian G-space whose moment map takes values in a G-bitorsor $\mathbb{G}$. The actions of G and Γ naturally induce an action of $G \rtimes \Gamma$ on M given by $(g, \varphi) \cdot p := g \cdot (\varphi \cdot p)$, $((g, \varphi) \in G \rtimes \Gamma,\ p \in M)$, and a moment map $\tilde{\mu} := M \to \mathbb{G} \rtimes \Gamma, p \mapsto (\mu(p), 1_\Gamma)$, where $\mathbb{G} \rtimes \Gamma$ is the bitorsor introduced in Remark 1.3. Then we obtain a twisted quasi-Hamiltonian $G \rtimes \Gamma$-space $(M, \omega, \tilde{\mu})$.

1.3. **The double $\mathbf{D}(\mathbb{G}_1, \mathbb{G}_2)$.** We give a basic example of a Γ-compatible quasi-Hamiltonian space. Let $\mathbb{G}_1$ and $\mathbb{G}_2$ be Γ-compatible G-bitorsors. We define a $G\times G$-action on $D := \mathbb{G}_1 \times \mathbb{G}_2$ by

$$(g_1, g_2) \cdot (a, b) := \big(g_1 a g_2^{-1}, g_2 b g_1^{-1}\big), \qquad (g_1, g_2) \in G \times G, \qquad (a, b) \in D.$$

Also we define a 2-form ω and a map μ by

$$\omega := -\frac{1}{2}\big(a^*\theta, b^*\overline{\theta}\big) - \frac{1}{2}\big(a^*\overline{\theta}, b^*\theta\big),$$
$$\mu\colon\ D \to \mathbb{G}_1 \cdot \mathbb{G}_2 \times \mathbb{G}_1^{-1} \cdot \mathbb{G}_2^{-1}, \qquad (a, b) \mapsto \big(ab, a^{-1}b^{-1}\big).$$

The tuple $\mathbf{D}(\mathbb{G}_1, \mathbb{G}_2) := (D, \omega, \mu)$ is a twisted quasi-Hamiltonian $(G \times G)$-space. Moreover, under the diagonal action of Γ on D and $\mathbb{G}_1\mathbb{G}_2 \times \mathbb{G}_1^{-1}\mathbb{G}_2^{-1}$, the space $\mathrm{D}(\mathbb{G}_1, \mathbb{G}_2)$ carries a Γ-compatible structure.

Remark 1.8. Let $\mathbb{C}$ and $\mathbb{H}$ be Γ-compatible G-bitorsors. We give the double $\mathbf{D}(\mathbb{C}, \mathbb{H})$ with different coordinates. For any $(C, h) \in \mathbf{D}(\mathbb{C}, \mathbb{H})$ and $(g, k) \in G \times G$, we define an action of $G \times G$ on $\mathbf{D}(\mathbb{C}, \mathbb{H})$ by $(g, k) \cdot (C, h) := \big(kCg^{-1}, khk^{-1}\big)$. We also define a quasi-Hamiltonian 2-form by

$$\omega := \frac{1}{2}\big(C^*\overline{\theta}, \mathrm{Ad}_h\, C^*\overline{\theta}\big) + \frac{1}{2}\big(C^*\overline{\theta}, h^*\theta + h^*\overline{\theta}\big),$$

and a moment map by

$$\mu\colon\ \mathbf{D}(\mathbb{C}, \mathbb{H}) \to \mathbb{C}^{-1}\mathbb{H}\mathbb{C} \times \mathbb{H}^{-1}, \qquad (C, h) \mapsto \big(C^{-1}hC, h^{-1}\big).$$

Then the twisted quasi-Hamiltonian $(G \times G)$-space $\mathbf{D}(\mathbb{C}, \mathbb{H})$ together with the natural Γ-action is also a Γ-compatible quasi-Hamiltonian $(G \times G)$-space and is isomorphic to the double of $\mathbb{C}^{-1}$ and $\mathbb{H}\mathbb{C}$ with standard coordinates.

2. Basic properties

Twisted quasi-Hamiltonian spaces have been investigated by several authors [5, 12, 15]. Although a Γ-compatible quasi-Hamiltonian G-space naturally gives rise to a twisted quasi-Hamiltonian $(G \rtimes \Gamma)$-space (see Remark 1.7), we study some basic properties of Γ-compatible quasi-Hamiltonian G-spaces, since this is the natural framework for the applications to twisted moduli spaces.

2.1. **The set of fixed points of (M, ω, μ).** We fix an action $\Gamma \to \mathrm{Aut}(G)$. Let (M, ω, μ) be a Γ-compatible q-Hamiltonian G-space with the moment map $\mu\colon M \to \mathbb{G}$. Let M^Γ be the fixed point set of M for the action of Γ and let $\mathbb{G}^\Gamma$ be the fixed point set of $\mathbb{G}$. We denote the identity component of G^Γ by H. Fix a connected component N of M^Γ, and we can then take the connected component $\mathbb{H}$ of $\mathbb{G}$ so that $\mu(N) \subset \mathbb{H}$. Then $\mathbb{H}$ becomes an H-bitorsor and the following theorem holds.

Theorem 2.1. *The tuple $(N, \omega|_N, \mu|_N)$ is a quasi-Hamiltonian H-space.*

Proof. Let θ^Γ (resp. $\overline{\theta}^\Gamma$) be the left (resp. right) Maurer–Cartan form on $\mathbb{H}$. We denote the inclusions by $\iota_N\colon N \to M$ and $\iota_{\mathbb{H}}\colon \mathbb{H} \to \mathbb{G}$. Let the Cartan 3-forms on $\mathbb{G}$ and $\mathbb{H}$ be $\chi := \frac{1}{12}(\theta, [\theta, \theta])_{\mathfrak{g}}$ and $\chi^\Gamma := \frac{1}{12}\big(\theta^\Gamma, \big[\theta^\Gamma, \theta^\Gamma\big]\big)_{\mathfrak{g}^\Gamma}$, respectively. Since $\chi^\Gamma = \chi|_{\mathbb{H}}$, we obtain the following $\mathrm{d}\omega|_N = \big(\mu^*\chi\big)|_N = (\mu|_N)^*\chi^\Gamma$.

Next, we will show the moment map condition. For any $\xi \in \mathfrak{g}^\Gamma$,

$$\iota\big(\xi^\#\big)\omega|_N = \big(\iota\big(\xi^\#\big)\omega\big)|_N = \left(\frac{1}{2}\mu^*\big(\theta + \overline{\theta}, \xi\big)_{\mathfrak{g}}\right)\Bigg|_N = \frac{1}{2}\mu|_N{}^*\big(\theta^\Gamma + \overline{\theta}^\Gamma, \xi\big)_{\mathfrak{g}^\Gamma}.$$

The degeneracy condition of $\omega|_N$ is shown by using the averaging over Γ. For any $x \in N$ and $X \in \ker \omega|_{N_x} \cap \ker \mathrm{d}_x\mu|_N$, $\mathrm{d}\mu \circ \mathrm{d}\iota_N(X) = \mathrm{d}\iota_{\mathbb{H}} \circ \mathrm{d}\mu|_N(X) = 0$. This

leads to $\mathrm{d}\iota_N(X) \in \ker \mathrm{d}_x\mu$. For any $Y \in T_xM$, the following identity holds from the Γ-invariance of ω

$$\omega(\mathrm{d}\iota_N(X), Y) = \frac{1}{|\Gamma|}\sum_{\varphi\in\Gamma} \varphi^*\omega(\mathrm{d}\iota_N(X), Y) = \omega\left(\mathrm{d}\iota_N(X), \frac{1}{|\Gamma|}\sum \varphi_* Y\right) = 0.$$

Therefore, we obtain $\mathrm{d}\iota_N(X) \in \ker \omega_x \cap \ker \mathrm{d}_x\mu = \{0\}$. So, we conclude

$$\ker \omega|_N \cap \ker \mathrm{d}\mu|_N = \{0\}. \qquad \square$$

2.2. **Fusion products and Γ-compatibilities.** Let G, H_1, and H_2 be compact Lie groups. Let $\mathbb{G}$, $\mathbb{H}_1$, and $\mathbb{H}_2$ be bitorsors for G, H_1, and H_2 respectively. We fix actions of a finite group Γ on G and H_j. Let (M_j, ω_j, μ_j) be a Γ-compatible quasi-Hamiltonian $(G \times H_j)$-space for $j = 1, 2$. We denote the moment map $\mu_j \colon M_j \to \mathbb{G}_j \times \mathbb{H}_j$ by $\mu_j = \big(\mu_j^G, \mu_j^H\big)$.

Proposition 2.2. *The fusion product $M_1 \circledast M_2$ carries a Γ-compatible quasi-Hamiltonian $G \times H_1 \times H_2$-structure for the diagonal action of Γ on $M_1 \circledast M_2$, $G_1 \times H_1 \times H_2$ and $\mathbb{G}_1\mathbb{G}_2 \times \mathbb{H}_1 \times \mathbb{H}_2$.*

Proof. We show the quasi-Hamiltonian 2-form on $M_1 \circledast M_2$, $\omega = \omega_1 + \omega_2 - \frac{1}{2}\big(\big(\mu_1^G\big)^*\theta, \big(\mu_2^G\big)^*\overline{\theta}\big)$ is Γ-invariant. Since $\mathbb{G}_j$ are Γ-compatible and the inner product $(\ ,\ )$ on $\mathfrak{g}$ is invariant under the action of Γ, for any $\varphi \in \Gamma$,

$$\begin{aligned}\varphi^*\omega &= \varphi^*\omega_1 + \varphi^*\omega_2 - \frac{1}{2}\varphi^*\big(\big(\mu_1^G\big)^*\theta, \big(\mu_2^G\big)^*\overline{\theta}\big) = \omega_1 + \omega_2 - \frac{1}{2}\big(\varphi\cdot\big(\mu_1^G\big)^*\theta, \varphi\cdot\big(\mu_2^G\big)^*\overline{\theta}\big)\\ &= \omega_1 + \omega_2 - \frac{1}{2}\big(\big(\mu_1^G\big)^*\theta, \big(\mu_2^G\big)^*\overline{\theta}\big) = \omega.\end{aligned}$$

The conditions (1) and (3) in Definition 1.6 are easy to verify in this case. $\square$

Moreover, suppose that M_j^Γ are non-empty connected sets and $\mathbb{G}_j^\Gamma$ and $\mathbb{H}_j^\Gamma$ are non-empty. One can easily prove the following proposition.

Proposition 2.3. *There exists a canonical twisted quasi-Hamiltonian isomorphism $M_1^\Gamma \circledast M_2^\Gamma \simeq (M_1 \circledast M_2)^\Gamma$.*

Remark 2.4. In Proposition 2.3, if M_j^Γ are not connected, we can obtain a similar result as above by taking connected components of M_1^Γ and M_2^Γ.

Example 2.5 (fused double $\mathbb{D}(\mathbb{G}_1, \mathbb{G}_2)$)**.** Let $\mathbb{G}_1$ and $\mathbb{G}_2$ be Γ-compatible G-bitorsors. The internal fusion of the double $\mathbf{D}(\mathbb{G}_1, \mathbb{G}_2)$ (see Example 1.3) is denoted by $\mathbb{D}(\mathbb{G}_1, \mathbb{G}_2)$, and is called the fused double. The space $\mathbb{D}(\mathbb{G}_1, \mathbb{G}_2)$ carries a Γ-compatible structure. Moreover, $\mathbb{G}_1^\Gamma$ and $\mathbb{G}_2^\Gamma$ are submanifolds of $\mathbb{G}_1$ and $\mathbb{G}_2$, respectively. We obtain an isomorphism of twisted quasi-Hamiltonian G^Γ-space $\mathbb{D}(\mathbb{G}_1, \mathbb{G}_2)^\Gamma \simeq \mathbb{D}\big(\mathbb{G}_1^\Gamma, \mathbb{G}_2^\Gamma\big)$.

3. Cyclic group actions on loop groups

The results of this section provide the foundation for the constructions in Section 4. We introduce the notion of Hamiltonian LG-spaces compatible with an action of a cyclic group $\Gamma := \mathbb{Z}/m\mathbb{Z}$, examine their relationship with Hamiltonian $L^{(\kappa)}G$-spaces, and discuss the equivalence theorem in the Γ-compatible setting. The constructions and computations involving the canonical 2-form on LG follow the same arguments as those in the (twisted) loop group setting developed in [2, 13].

3.1. **Twisted loop groups.** Let G be a compact connected simply connected Lie group with Lie algebra $\mathfrak{g}$. We fix an Ad-invariant inner product $(\ ,\ )$ on $\mathfrak{g}$ and a Lie group automorphism $\kappa \in \mathrm{Aut}(G)$ such that there exists a positive integer m such that $\kappa^m = 1$. We suppose that κ preserves the inner product $(\ ,\ )$ on $\mathfrak{g}$. Let S^1_m denote $\mathbb{R}/m\mathbb{Z}$. We define the loop group LG as a space of maps

$$LG := \mathrm{Map}\big(S^1_m, G\big)$$

of a fixed Sobolev class $\lambda > 1/2$. Also we define the κ-twisted loop group $L^{(\kappa)}G$ as a subgroup of LG

$$L^{(\kappa)}G := \big\{ g \in LG \mid \kappa g(t) = g(t+1) \text{ for all } t \in S^1_m \big\}.$$

We define the spaces $L^{(\kappa)}\mathfrak{g}$ and $L^{(\kappa)}\mathfrak{g}^*$ as subspaces of the space of maps $L\mathfrak{g} := \mathrm{Map}\big(S^1_m, \mathfrak{g}\big)$ of Sobolev class λ and the space of $\mathfrak{g}$-valued 1-forms $L\mathfrak{g}^* := \Omega^1\big(S^1_m, \mathfrak{g}\big)$ of Sobolev class $\lambda - 1$,

$$L^{(\kappa)}\mathfrak{g} := \big\{ \xi \in L\mathfrak{g} \mid \kappa\xi(t) = \xi(t+1) \text{ for all } t \in S^1_m \big\},$$
$$L^{(\kappa)}\mathfrak{g}^* := \big\{ A \in L\mathfrak{g}^* \mid \kappa A_t = A_{t+1} \text{ for all } t \in S^1_m \big\}.$$

The gauge transformation on $L^{(\kappa)}\mathfrak{g}^*$ is defined by $g \cdot A := gAg^{-1} + \mathrm{d}gg^{-1}$, $g \in L^{(\kappa)}G$, $A \in L^{(\kappa)}\mathfrak{g}^*$. There exists a canonical pairing $L^{(\kappa)}\mathfrak{g}^* \times L^{(\kappa)}\mathfrak{g} \to \mathbb{R}$, $(A, \xi) \mapsto \int_0^1 (A, \xi)$. We define the holonomy $\mathrm{Hol}^b_t(A)$ of A starting at $b \in S^1_m$ by the unique solution of the following differential equation:

$$\mathrm{Hol}^b_t(A)^{-1} \frac{\partial \mathrm{Hol}^b_t(A)}{\partial t} = -A, \qquad \mathrm{Hol}^b_b(A) = 1_G.$$

Then, the holonomy and the gauge action satisfy the following equation:

$$\mathrm{Hol}^b_t(g \cdot A) = g(b)\,\mathrm{Hol}^b_t(A) g(t)^{-1}, \qquad g \in L^{(\kappa)}G, \qquad A \in L^{(\kappa)}\mathfrak{g}^*.$$

Therefore, we have the holonomy map $\mathrm{Hol}^{(\kappa)} \colon L^{(\kappa)}\mathfrak{g}^* \to G^{(\kappa)}$, $A \mapsto \mathrm{Hol}^0_1(A)$, where $G^{(\kappa)} = G$ is a G-bitorsor with the following left and right actions $g \cdot x := gx$, $x \cdot g := x(\kappa g)$, $\big(x \in G^{(\kappa)},\ g \in G\big)$.

Let θ (resp. $\overline{\theta}$) be the left (resp. right) Maurer–Cartan form on G. We define the 2-form on $L^{(\kappa)}\mathfrak{g}^*$ by

$$\varpi^{(\kappa)} := \frac{1}{2} \int_0^1 \mathrm{d}s \left(\mathrm{Hol}^{0\,*}_s \overline{\theta}, \frac{\partial}{\partial s} \mathrm{Hol}^{0\,*}_s \overline{\theta} \right).$$

The following proposition is analogous to [2, Proposition 8.1].

Proposition 3.1. *We have*

$$\varpi^{(\kappa)} \text{ is } L^{(\kappa)}G\text{-invariant}, \qquad \mathrm{d}\varpi^{(\kappa)} = -{\mathrm{Hol}^{(\kappa)}}^* \frac{1}{12}(\theta, [\theta, \theta]),$$
$$\iota\big(\xi^{\#}\big)\varpi^{(\kappa)} = -\mathrm{d} \int_0^1 (A, \xi) - \frac{1}{2} {\mathrm{Hol}^{(\kappa)}}^* \big(\theta + \overline{\theta}, \xi\big), \qquad \xi \in L^{(\kappa)}\mathfrak{g}.$$

Proof. This is shown in the same way as in Proposition 3.2. □

3.2. **Cyclic group actions.** In this subsection, let Γ be a cyclic group $\mathbb{Z}/m\mathbb{Z}$, and let S^1 denote $\mathbb{R}/m\mathbb{Z}$. We consider a Γ-action on G, i.e., a group homomorphism $\Gamma \to \mathrm{Aut}(G)$. Also a natural Γ-action on S^1 is defined by $t \cdot l := t - l$ for $t \in S^1$ and $l \in \Gamma$. Then we can define a Γ-action on the loop group LG by

$$l \cdot g(t) := \kappa^l g(t \cdot l), \qquad g \in LG, \qquad t \in S^1, \qquad l \in \Gamma.$$

Γ-actions on $L\mathfrak{g}$ and $L\mathfrak{g}^*$ are defined in the same way. Let β be a finite subset $0 \cdot \Gamma = \{0, 1, \ldots, m-1\} \subset S^1$, and we denote $G^\beta := \mathrm{Map}(\beta, G)$. Also we denote Maurer–Cartan forms on G^β by θ and $\overline{\theta}$. Similar to the twisted case described above,

gauge transformations, holonomies and natural pairing are defined. Furthermore, we define the 2-form on $L\mathfrak{g}^*$ by

$$\varpi := \frac{1}{2}\sum_{i=1}^{m}\int_i^{i+1} \mathrm{d}s \left({\mathrm{Hol}_s^i}^* \overline{\theta}, \frac{\partial}{\partial s} {\mathrm{Hol}_s^i}^* \overline{\theta} \right).$$

Also the holonomy map $\mathrm{Hol}\colon L\mathfrak{g}^* \to \mathbb{H} := G^\beta$ is defined by $\mathrm{Hol}(A) := \big(\mathrm{Hol}_{i+1}^i(A)\big)_{i=1}^m$, $\big(A \in L\mathfrak{g}^*\big)$, where $\mathbb{H}$ is a Γ-compatible G^β-bitorsor (Example 1.2).

In this case, the following proposition holds in the same way as in [2, Proposition 8.1].

Proposition 3.2. *We have*

$$\varpi \text{ is } LG\text{-invariant}, \qquad \mathrm{d}\varpi = -\mathrm{Hol}^* \frac{1}{12}(\theta, [\theta, \theta]),$$
$$\iota\big(\xi^\#\big)\varpi_A = -\mathrm{d}\oint (A, \xi) - \frac{1}{2}\mathrm{Hol}^*\big(\theta + \overline{\theta}, \xi|_\beta\big), \qquad \xi \in L\mathfrak{g}, \qquad A \in L\mathfrak{g}^*,$$

where $\xi|_\beta = (\xi(i))_{i\in\beta} \in \mathfrak{g}^\beta$.

To prove it, we use the following lemma.

Lemma 3.3. *We have*

$$\iota\big(\xi^\#\big){\mathrm{Hol}_s^i}^*\overline{\theta} = \mathrm{Ad}\big(\mathrm{Hol}_s^i\big)\xi(s) - \xi(i), \qquad \xi \in L\mathfrak{g}.$$
$$\iota(\eta){\mathrm{Hol}_s^i}^*\overline{\theta} = -\int_i^s \mathrm{d}u\, \mathrm{Ad}\big(\mathrm{Hol}_u^i\big)\eta, \qquad \eta \in T_A L\mathfrak{g}^* = L\mathfrak{g}^*.$$

3.3. **Γ-compatibilities of Hamiltonian LG-spaces.** In this subsection, we state the definition of a Γ-compatibility on Hamiltonian LG-spaces. This is the usual Hamiltonian space version of a Γ-compatibility.

First, we recall the definition of a Hamiltonian LG-space [2].

Definition 3.4. A Hamiltonian LG-space is a Banach-manifold N together with an LG-action, an invariant 2-form $\sigma \in \Omega^2(N)^{LG}$, and an equivariant map $\Phi \in C^\infty\big(N, L\mathfrak{g}^*\big)^{LG}$ such that

(1) The 2-form σ is closed.
(2) The map Φ is a moment map for the LG-action $\iota\big(\xi^\#\big)\sigma = -\mathrm{d}\oint_{S^1}(m, \xi)$.
(3) The 2-form σ is weakly non-degenerate, that is, the induced map $\sigma_x^\flat\colon T_xN \to T_x^*N$ is injective.

Definition 3.5. Let (N, σ, Φ) be a Hamiltonian LG-space and we fix a Γ-action on N. The tuple (N, σ, Φ) together with the Γ-action is called a Γ-compatible Hamiltonian LG-space if it satisfies the following properties:

(1) $l \cdot (g \cdot x) = (l \cdot g) \cdot (l \cdot x)$, $l \in \Gamma$, $g \in LG$, $x \in N$.
(2) σ is Γ-invariant.
(3) $\Phi\colon N \to L\mathfrak{g}^*$ is Γ-equivariant.

We say that two Γ-compatible Hamiltonian LG-spaces are isomorphic if there exists a Γ-equivariant isomorphism of Hamiltonian LG-spaces.

3.4. **Fixed points of a Γ-compatible space.** Let (N, σ, Φ) be a Γ-compatible Hamiltonian LG-space. Then, we obtain a Hamiltonian $L^{(\kappa)}G$-space from a Γ-compatible Hamiltonian LG-space by taking fixed point objects. Now Γ-fixed objects are defined by

- $N^\Gamma := \{x \in N \mid l \cdot x = x,\ l \in \Gamma\}$,
- $\sigma^\Gamma := \sigma|_{N^\Gamma}$,
- $\Phi^\Gamma\colon N^\Gamma \to L\mathfrak{g}^*$, $x \mapsto \Phi(x)$.

In this paper, we denote Γ-fixed objects as above. For example, the Γ-fixed subgroup of LG is denoted by LG^Γ. Each connected component of N^Γ is a Banach submanifold of N. By considering left transformation of LG^Γ, the subgroup LG^Γ is a Banach Lie subgroup of LG (see Appendix 6).

We thus obtain the following proposition.

Proposition 3.6. *For any connected component L of N^Γ, the tuple $\big(L, \sigma^\Gamma|_L, \Phi^\Gamma|_L\big)$ is a Hamiltonian LG^Γ-space, i.e., the following properties hold:*

- *LG^Γ acts on L.*
- *$\sigma^\Gamma|_L$ is a LG^Γ-invariant symplectic form on L.*
- *$\Phi^\Gamma|_L$ is LG^Γ-equivariant and $\sigma^\Gamma|_L$, $\Phi^\Gamma|_L$ satisfy the moment map condition*

$$\iota\big(\xi^\#\big)\sigma^\Gamma = -\mathrm{d}\oint\big(\Phi^\Gamma, \xi\big), \qquad \xi \in L\mathfrak{g}^\Gamma.$$

Proof. Since G is simply connected, the space LG^Γ is connected. Therefore, LG^Γ acts on a connected component L. The moment map condition for $\Phi^\Gamma|_L$ is a straightforward calculation and the weak non-degeneracy of σ^Γ is proved by averaging over Γ. Let $\iota\colon N^\Gamma \to N$ be the inclusion map. For any $\xi \in L\mathfrak{g}^\Gamma$,

$$\iota\big(\xi^\#\big)\sigma^\Gamma|_L = \big(\iota\big(\xi^\#\big)\sigma\big)|_L = -\mathrm{d}\oint(\Phi, \xi)\Big|_L = -\oint\big(\Phi^\Gamma|_L, \xi\big).$$

Next, we will show the non-degeneracy of σ^Γ. Fix an element $x \in N^\Gamma$ and $X \in \ker\sigma^\Gamma$. Then for any $Y \in T_xN$,

$$\iota(Y)\iota(X)\sigma = \sigma(X,Y) = \frac{1}{|\Gamma|}\sum_{l\in\Gamma} l^*\sigma(X,Y) = \sigma\left(X,\ \frac{1}{|\Gamma|}\sum_{l\in\Gamma} l_*Y\right) = 0.$$

Thus we have proved the non-degeneracy condition. □

Remark 3.7. Suppose that Γ is a cyclic group $\mathbb{Z}/m\mathbb{Z}$ and that the action of Γ on LG is the one defined in Section 3.1. Then we can identify LG^Γ with the twisted loop group $L^{(\kappa)}G$,

$$LG^\Gamma = \{g \in LG \mid l\cdot g = g,\ l \in \Gamma\} = \big\{g \in LG \mid \kappa g(t) = g(t+1) \text{ for all } t \in S^1_m\big\} = L^{(\kappa)}G.$$

For any $\xi \in L^{(\kappa)}\mathfrak{g} = L\mathfrak{g}^\Gamma$,

$$\iota\big(\xi^\#\big)\sigma^\Gamma = -\mathrm{d}\oint\big(\Phi^\Gamma, \xi\big) = -\mathrm{d}\sum_{i=1}^{m}\int_i^{i+1}\big(\Phi^\Gamma, \xi\big) = -m\mathrm{d}\int_0^1(\Phi, \xi).$$

Therefore, $\big(L, \frac{1}{m}\sigma^\Gamma, \Phi^{(\kappa)}\big)$ becomes a Hamiltonian $L^{(\kappa)}G$-space.

3.5. **Equivalence theorem.** It is known that there exists a natural correspondence between (twisted) Hamiltonian LG-spaces and (twisted) quasi-Hamiltonian G-spaces which is called the equivalence theorem [2, 13, 15]. Taking base points β with $\beta = 0\cdot\Gamma \subset S^1_m$, the equivalence theorem between Γ-compatible Hamiltonian LG-spaces and Γ-compatible quasi-Hamiltonian G^β-spaces can be shown. Let $\Omega^\beta G$ be a based loop group for β, i.e., $\Omega^\beta G = \{g \in LG \mid g|_\beta = 1_G\}$.

Theorem 3.8. *Taking the holonomy manifold for $\Omega^\beta G$ yields a bijective correspondence between*

- *the isomorphism class of Γ-compatible Hamiltonian LG-spaces (N, σ, Φ) with a proper moment map,*
- *the isomorphism class of Γ-compatible q-Hamiltonian G^β-spaces (M, ω, μ).*

4. Twisted connections and finite group actions

4.1. **Preliminaries.** Let Σ be a compact connected 2-manifold with boundary components $V_0, V_1, \dots, V_r$ and $P := \Sigma \times G$ a trivial bundle on Σ. Fix $\lambda > 1$ and let $\Omega^j(\Sigma, \mathfrak{g})$ be the space of $\mathfrak{g}$-valued j-forms of Sobolev class $\lambda - j$. Then the gauge $\mathcal{G}(\Sigma) := \mathrm{Map}(\Sigma, G)$ is a Banach Lie group modeled on $\mathrm{Lie}(\mathcal{G}(\Sigma)) = \Omega^0(\Sigma, \mathfrak{g})$. Let $\mathcal{A}(\Sigma) := \Omega^1(\Sigma, \mathfrak{g})$ be the space of connections on the trivial bundle on Σ. $\mathcal{A}(\Sigma)$ has a smooth $\mathcal{G}(\Sigma)$-action given by $g \cdot A = gAg^{-1} + g^*\overline{\theta}$ The space $\mathcal{A}(\Sigma)$ has a canonical symplectic form called Atiyah–Bott 2-form $\sigma = \frac{1}{2}\int_\Sigma (\delta A \wedge \delta A)$ for which $\mathcal{G}(\Sigma)$-action is Hamiltonian with moment map

$$\langle \Psi(A), \xi \rangle = \int_\Sigma (\mathrm{curv}\, A, \xi) - \int_{\partial\Sigma} (A, \xi), \qquad \xi \in \Omega^0(\Sigma, \mathfrak{g}),$$

where $\mathrm{curv}(A)$ is the curvature of A. The space of flat connections is denoted by $\mathcal{A}_{\mathrm{flat}}(\Sigma)$.

4.2. **Finite group actions.** Let Γ be a finite group. Fix a free right Γ-action on Σ which preserves the orientation of Σ, and a left Γ-action on G which preserves the inner product on $\mathfrak{g}$. Then a right Γ-action on the trivial bundle is defined by

$$(x, a) \cdot \varphi = \big(x \cdot \varphi, \varphi^{-1} \cdot a\big), \qquad (x, a) \in P, \qquad \varphi \in \Gamma.$$

This action satisfies Γ-compatibility

$$(p \cdot g) \cdot \varphi = (p \cdot \varphi) \cdot (g \cdot \varphi), \qquad p \in P, \qquad g \in G, \qquad \varphi \in \Gamma,$$

where $g \cdot \varphi = \varphi^{-1} \cdot g$. We can also define left Γ-actions on $\Omega^j(P, \mathfrak{g})$ by

$$\varphi \cdot \alpha = \varphi \circ \alpha \circ d\varphi, \qquad \alpha \in \Omega^j(P, \mathfrak{g}), \qquad \varphi \in \Gamma,$$

and left Γ-actions on $\Omega^j(\Sigma, \mathfrak{g})$ can be defined in the same way.

Lemma 4.1. *The canonical 2-form σ is Γ-invariant and the curvature* $\mathrm{curv}\colon \mathcal{A}(\Sigma) \to \Omega^2(\Sigma, \mathfrak{g})$ *is Γ-equivariant. Moreover, the following Γ-compatibility holds:*

$$\varphi \cdot (g \cdot A) = (\varphi \cdot g) \cdot (\varphi \cdot A), \qquad \varphi \in \Gamma, \qquad g \in \mathcal{G}(\Sigma), \qquad A \in \mathcal{A}(\Sigma).$$

Proof. For any $\varphi \in \Gamma$ and $X, Y \in T_A\mathcal{A}(\Sigma) = \Omega^1(\Sigma, \mathfrak{g})$, the following holds:

$$\varphi^*\sigma(X, Y) = \int_\Sigma (\varphi X \circ \mathrm{d}\varphi \wedge \varphi Y \circ \mathrm{d}\varphi) = \int_\Sigma (X \wedge Y) = \sigma(X, Y).$$

Since the exterior derivative $\mathrm{d}\colon \Omega^j(\Sigma, \mathfrak{g}) \to \Omega^{j+1}(\Sigma, \mathfrak{g})$ is Γ-equivariant, the Γ-equivariance of the curvature holds

$$\mathrm{curv}(\varphi \cdot A) = \mathrm{d}(\varphi \cdot A) + \frac{1}{2}[\varphi \cdot A \wedge \varphi \cdot A] = \varphi \cdot \mathrm{d}A + \frac{1}{2}\varphi \cdot [A \wedge A] = \varphi \cdot \mathrm{curv}\, A,$$

where $\varphi \in \Gamma$ and $A \in \mathcal{A}(\Sigma)$. Moreover, for any $g \in \mathcal{G}(\Sigma)$, $\varphi \in \Gamma$ and $A \in \mathcal{A}(\Sigma)$, the following holds $\varphi \circ \mathrm{Ad}(g) = \mathrm{Ad}(\varphi \cdot g) \circ \varphi$, $\varphi \cdot g^*\overline{\theta} = (\varphi \cdot g)^*\overline{\theta}$. Therefore, we obtain the Γ-compatibility of $\mathcal{G}(\Sigma)$ and $\mathcal{A}(\Sigma)$. □

Γ-actions on $\mathcal{G}(\partial\Sigma)$ and $\Omega^j(\partial\Sigma, \mathfrak{g})$ are also defined in the same way.

4.3. **The space $\mathcal{M}(\Sigma)$ and Γ-compatibilities.** Since the restriction map $\mathcal{G}(\Sigma) \to \mathcal{G}(\partial\Sigma)$ is surjective, the following sequence is exact:

$$1 \to \mathcal{G}_{\partial\Sigma}(\Sigma) \to \mathcal{G}(\Sigma) \to \mathcal{G}(\partial\Sigma) \to 1,$$

where $\mathcal{G}_{\partial\Sigma}(\Sigma) := \{g \in \mathcal{G}(\Sigma) \mid g|_{\partial\Sigma} = 1_G\}$. Then $\mathcal{M}(\Sigma) := \mathcal{A}_{\mathrm{flat}}(\Sigma)/\mathcal{G}_{\partial\Sigma}(\Sigma)$ is a Banach manifold and a Hamiltonian $(\mathcal{G}(\partial\Sigma) \simeq LG^{r+1})$-space with proper moment map [2, 8]. The symplectic form σ on $\mathcal{M}(\Sigma)$ is induced from Atiyah–Bott 2-form on $\mathcal{A}(\Sigma)$ and the moment map is a restriction

$$\Phi\colon \mathcal{M}(\Sigma) \to \Omega^1(\partial\Sigma, \mathfrak{g}), \qquad A \mapsto A|_{\partial\Sigma}.$$

The Γ-compatibility of $\mathcal{M}(\Sigma)$ can be shown by using Lemma 4.1. A Γ-action on $\mathcal{M}(\Sigma)$ is defined by $\varphi \cdot [A] := [\varphi \cdot A]$, $\varphi \in \Gamma$, $[A] \in \mathcal{M}(\Sigma)$. The Hamiltonian $\mathcal{G}(\partial\Sigma)$-space $(\mathcal{M}(\Sigma), \sigma, \Phi)$ together with the Γ-action is a Γ-compatible Hamiltonian $\mathcal{G}(\partial\Sigma)$-space.

Proposition 4.2. *$(\mathcal{M}(\Sigma), \sigma, \Phi)$ is a Γ-compatible Hamiltonian $\mathcal{G}(\partial\Sigma)$-space.*

The above proposition follows immediately from Lemma 4.1.

4.4. **Fixed points of $\boldsymbol{\mathcal{M}(\Sigma)}$.** By taking a fixed point objects $\big(\mathcal{M}(\Sigma)^\Gamma, \omega^\Gamma, \Phi^\Gamma\big)$, we get Hamiltonian spaces for twisted loop groups actions.

We define a Γ-action on $I := \{0, 1, \dots, r\}$ by using a permutation of $\partial\Sigma = \coprod_{i=0}^r V_i$, $V_{i\cdot\varphi} = V_i \cdot \varphi$. Let $\Gamma(i)$ be a stabilizer of $i \in I$, i.e., $\Gamma(i) = \{\varphi \in \Gamma \mid i \cdot \varphi = i\}$.

Lemma 4.3. *For any $i \in I$, $\Gamma(i)$ is a cyclic group.*

Proof. Fix a point $b \in V_i$ and consider a $\Gamma(i)$-orbit $b\cdot\Gamma(i)$, we restrict the $\Gamma(i)$-action on $\partial\Sigma$ to $b\cdot\Gamma(i)$. Since this Γ-action on $\partial\Sigma$ is free, the restricted $\Gamma(i)$-action on $b\cdot\Gamma(i)$ is simply transitive. So by fixing a cyclic order of $b\cdot\Gamma(i) = \{b_0, b_1, \dots, b_{m_i-1}\}$, we get a unique element $b_1 = b_0 \cdot \kappa_i$. Then, the element κ_i is a generator of $\Gamma(i) = \langle\kappa_i\rangle$. □

We fix a generator κ_i of $\Gamma(i)$ for all $i \in I$. We set $m_i := |\Gamma(i)|$, and identify V_i with $S^1_{m_i} = \mathbb{R}/m_i\mathbb{Z}$ while keeping $\Gamma(i)$-action. So, the following isomorphism is obtained: $\mathcal{G}(V_i)^{\Gamma(i)} \simeq L^{(\kappa_i)}G$. So, by taking a complete system of representatives I' of I/Γ, and fixing κ_i for any $i \in I$, and taking $\tau_{ji} \in \Gamma$ such that $\kappa_j = \tau_{ji}\kappa_i\tau_{ji}^{-1}$ for $i, j \in i \cdot \Gamma$, we identify the fixed points subgroup $\mathcal{G}(\partial\Sigma)^\Gamma$ with a product of twisted loop groups,

$$\mathcal{G}(\partial\Sigma)^\Gamma \simeq \left\{ (g_i) \in \prod_{i\in I} L^{(\kappa_i)}G \,\middle|\, g_i = \tau_{ij} g_j \right\} \simeq \prod_{i\in I'} L^{(\kappa_i)}G.$$

In the same way, we obtain the following isomorphisms:

$$\Omega^1(\partial\Sigma, \mathfrak{g})^\Gamma \simeq \prod_{i\in I'} L^{(\kappa_i)}\mathfrak{g}^*; \qquad A \mapsto (A|_{V_i}), \qquad \Omega^0(\partial\Sigma, \mathfrak{g})^\Gamma \simeq \prod_{i\in I'} L^{(\kappa_i)}\mathfrak{g}; \qquad \xi \mapsto (\xi|_{V_i}).$$

We define Φ' and σ' by $\Phi'\colon \mathcal{M}(\Sigma)^\Gamma \to \prod_{i\in I'} L^{(\kappa_i)}\mathfrak{g}^*$; $A \mapsto (\Phi(A)|_{V_i})$, $\sigma' := \frac{1}{|\Gamma|}\sigma$. By considering the $\prod_{i\in I} L^{(\kappa_i)}G$-action on $\mathcal{M}(X)$, the following proposition holds.

Proposition 4.4. *Each connected component $\mathcal{N}$ of $\mathcal{M}(\Sigma)^\Gamma$ together with the 2-form $\sigma'|_{\mathcal{N}}$ and the moment map $\Phi'|_{\mathcal{N}}$ is a Hamiltonian $\Pi_{i\in I'} L^{(\kappa_i)}G$-space.*

Proof. It is trivial that σ' is symplectic and Φ' is equivariant for twisted loop group actions. So, we prove the moment map condition: for any $[A] \in \mathcal{M}(\Sigma)$ and $(\xi_i) = \xi \in \prod_{i\in I'} L^{(\kappa_i)}\mathfrak{g}$,

$$\begin{aligned}
\iota\big(\xi^\#\big)\sigma'_{[A]} &= \frac{1}{|\Gamma|}\iota\big(\xi^\#\big)\sigma_{[A]}|_\Gamma = -\frac{1}{|\Gamma|}\mathrm{d}\int_{\partial\Sigma}(\Phi(A), \xi) = -\frac{1}{|\Gamma|}\sum_{i\in I}\mathrm{d}\oint_{V_i}(A|_{V_i}, \xi|_{V_i}) \\
&= -\frac{1}{|\Gamma|}\sum_{i\in I'}|i\cdot\Gamma|\,\mathrm{d}\oint_{V_i}(A|_{V_i}, \xi_i) = -\frac{1}{|\Gamma|}\sum_{i\in I'}|i\cdot\Gamma|\,|\Gamma(i)|\,\mathrm{d}\int_0^1 (A|_{V_i}, \xi_i) \\
&= -\sum_{i\in I'}\mathrm{d}\int_0^1 \big(\Phi'(A)_i, \xi_i\big).
\end{aligned}$$

□

4.5. **The space $M(\Sigma)$ and Γ-compatibilities.** Let $\beta \subset \partial\Sigma$ be a finite subset (base points) which is Γ-invariant and we fix a cyclic ordering of $\beta_i = \{b_\lambda^i\} := \beta \cap V_i$. We denote $\mathcal{G}_\beta(\Sigma) := \{g \in \mathcal{G}(\Sigma) \mid g|_\beta = 1\}$, $\mathcal{G}_\beta(\partial\Sigma) := \{g \in \mathcal{G}(\partial\Sigma) \mid g|_\beta = 1\}$. Set $M(\Sigma) := \mathcal{A}_{\mathrm{flat}}(\Sigma)/\mathcal{G}_\beta(\Sigma) \simeq \mathcal{M}(\Sigma)/\mathcal{G}_\beta(\partial\Sigma)$. An action of $\mathcal{G}(\partial\Sigma)/\mathcal{G}_\beta(\partial\Sigma) \simeq G^\beta$ on $M(\Sigma)$ is induced from the residual gauge action on $\mathcal{M}(\Sigma)$. The collection of holonomy map $\mathrm{Hol}_{b_{\lambda+1}^i}^{b_\lambda^i}(A)$ defines a G^β-equivariant map

$$\mu\colon\ M(\Sigma) \to \mathbb{H}, \qquad A \mapsto \mathrm{Hol}(A) := \big(\mathrm{Hol}_{b_{\lambda+1}^i}^{b_\lambda^i}(A)\big)_{b_\lambda^i \in \beta},$$

where $\mathbb{H} = G^\beta$ is a G^β-bitorsor under the following left and right actions, $(g \cdot h)(b_\lambda^i) = g(b_\lambda^i)h(b_\lambda^i)$, $(h \cdot g)(b_\lambda^i) = h(b_\lambda^i)g(b_{\lambda+1}^i)$, $g \in G^\beta$, $h \in \mathbb{H}$. An action of Γ on $M(\Sigma)$ is induced from the Γ-action on $\mathcal{M}(\Sigma)$. The following theorem, which is analogous to [2, Theorem 9.1], holds.

Theorem 4.5. *There is a natural G^β-invariant 2-form ω on $M(\Sigma)$ for which $(M(\Sigma), \omega, \mu)$ is a Γ-compatible q-Hamiltonian G^β-space.*

Proof. The proof is analogous to that of [2, Theorem 9.1]. In this proof, let $\pi\colon \mathcal{M}(\Sigma) \to M(\Sigma)$ be the quotient map. First, we will show the equivariance of $\mu\colon M(\Sigma) \to \mathbb{H}$. For any $g \in G^\beta$ and $\pi(A) \in M(\Sigma)$, $\mu(g \cdot \pi(A)) = \mu(\pi(\tilde{g} \cdot A)) = \mathrm{Hol}(\tilde{g} \cdot A) = \mathrm{Ad}_g\, \mu(\pi(A))$.

Let $R_j\colon \Omega^1(\partial\Sigma, \mathfrak{g}) \to \Omega^1(V_j, \mathfrak{g})$ be a restriction and let ϖ_j be the canonical 2-form on $\Omega^1(V_j, \mathfrak{g}) \simeq L\mathfrak{g}^*$. Denote $\varpi := \sum_{j=0}^r R_j^* \varpi_j$. We define a 2-form on $\mathcal{M}(\Sigma)$, $\tilde{\omega} := \sigma - \Phi^*\varpi$. Then the 2-form $\tilde{\omega}$ is basic for $\mathcal{G}_\beta(\partial\Sigma)$-action for $\mathcal{M}(\Sigma)$. Since σ and ϖ are $\mathcal{G}(\partial\Sigma)$-invariant, $\tilde{\omega}$ becomes a $\mathcal{G}(\partial\Sigma)$-invariant 2-form. Furthermore, for any $\xi \in \mathrm{Lie}(\mathcal{G}_\beta(\partial\Sigma))$,

$$\begin{aligned}\iota(\xi^\#)\tilde{\omega}_A &= \iota(\xi^\#)\sigma_A - \iota(\xi^\#)\Phi^*\varpi_A = \iota(\xi^\#)\sigma_A - \sum_{j=0}^r R_j{}^*\big(\iota(\xi|_{V_j}{}^\#)\varpi_j\big)\\ &= -\mathrm{d}\int_{\partial\Sigma}(A|_{\partial\Sigma}, \xi) - \sum_{j=0}^r R_j{}^*\left(-\mathrm{d}\oint(A|_{V_j}, \xi|_{V_j}) - \frac{1}{2}(\mathrm{Hol}_j)^*\big(\theta_j + \overline{\theta}_j, \xi|_{\beta_j}\big)\right) = 0,\end{aligned}$$

where $\mathrm{Hol}_j\colon \Omega^1(V_j, \mathfrak{g}) \to \mathbb{H}^j := G^{\beta_j}$, $A_j \mapsto \big(\mathrm{Hol}_{b_{\lambda+1}^j}^{b_\lambda^j}(A_j)\big)$, and θ_j (resp. $\overline{\theta}$) is the left (resp. right) Maurer–Cartan form on $\mathbb{H}^j$. So, $\tilde{\omega}$ induces a unique G^β-invariant 2-form ω on $M(\Sigma)$ which satisfies $\pi^*\omega = \tilde{\omega}$.

Now we prove the tuple $(M(\Sigma), \omega, \mu)$ is a twisted quasi-Hamiltonian G^β-space. We denote the canonical 3-form on $\mathbb{H}$ by $\chi_{\mathbb{H}}$, i.e., $\chi_{\mathbb{H}} = \frac{1}{12}(\theta_{\mathbb{H}}, [\theta_{\mathbb{H}}, \theta_{\mathbb{H}}])$. Then, by $\mathrm{Hol} \circ \Phi = \mu \circ \pi$, $\pi^*\mathrm{d}\omega = \mathrm{d}\tilde{\omega} = \mathrm{d}(\sigma - \Phi^*\varpi) = \Phi^* \mathrm{Hol}^*\, \chi_{\mathbb{H}} = \pi^*\mu^*\chi_{\mathbb{H}}$. Therefore, we have proved $\mathrm{d}\omega = \mu^*\chi_{\mathbb{H}}$.

Next, we want to show the moment map condition. For any $\xi \in \mathfrak{g}^\beta$, there exists $\tilde{\xi} \in \Omega^0(\partial\Sigma, \mathfrak{g})$ such that $\tilde{\xi}|_\beta = \xi$. Then

$$\begin{aligned}\pi^*\big(\iota(\xi^\#)\omega\big) = \iota(\tilde{\xi})\pi^*\omega = \iota(\tilde{\xi}^\#)(\sigma - \Phi^*\varpi) &= \frac{1}{2}\sum_{j=0}^r R_j{}^*\mathrm{Hol}_j{}^*\big(\theta_j + \overline{\theta}_j, \tilde{\xi}|_{\beta_j}\big)\\ &= \pi^*\left(\frac{1}{2}\mu^*\big(\theta_{\mathbb{H}} + \overline{\theta}_{\mathbb{H}}, \xi\big)\right).\end{aligned}$$

So, the moment map condition holds.

Finally, we will prove the degeneracy of ω, i.e., we will show $\ker\omega_m \cap \ker d_m\mu = \{0\}$ for any $m \in M(\Sigma)$. Fix $v \in \ker\omega_m \cap \ker \mathrm{d}_m\mu$ and $A \in \mathcal{M}(\Sigma)$ such that $\pi(A) = m$. Since $\pi\colon \mathcal{M}(\Sigma) \to M(\Sigma)$ is a submersion, there exists a tangent vector $X \in T_A\mathcal{M}(\Sigma)$ such that $\mathrm{d}_A\pi(X) = v$.

For any $Y \in T_A\mathcal{M}(\Sigma)$, $\tilde{\omega}_A(X, Y) = \pi^*\omega_A(X, Y) = 0$. Therefore, $X \in \ker\tilde{\omega}_A$. By the definition of $\tilde{\omega}$, the following holds: $\iota(X)\sigma = \iota(X)\Phi^*\varpi$. Thus, $X \in (\ker \mathrm{d}_A\Phi)^\sigma$.

Since the space $(\ker \mathrm{d}_A\Phi)^\sigma$ equals to the tangent space of the orbit $\mathcal{G}(\partial\Sigma)\cdot A$, i.e., there exists $\xi \in \Omega^0(\partial\Sigma,\mathfrak{g})$ that the generating vector $\xi^\#$ equals to X [2, Theorem 9.1].

Since ${\xi|_\beta}^\# \in \ker \mathrm{d}_m\mu$, we obtain $\xi|_\beta \in \ker(\mathrm{Ad}_{\mu(m)}-1)$. Now we define $\eta_j \in \Omega^0(V_j,\mathfrak{g})$ by $\eta_j(s) := \mathrm{Ad}\big(\mathrm{Hol}_s^{b_0^j}(\Phi(A))^{-1}\big)\xi\big(b_0^j\big)$. Because $\mathrm{Ad}_{\mu(m)}\,\xi|_\beta = \xi|_\beta$,

$$\eta_j\big(b_{\lambda_j}^j\big) = \mathrm{Ad}\big(\mathrm{Hol}_{b_{\lambda_j}^j}^{b_0^j}(\Phi(A))\big)\xi\big(b_0^j\big) = \xi\big(b_{\lambda_j}^j\big).$$

Therefore, η_j is well defined. By collecting η_j for all $j=0,\dots,r$, we define $\eta \in \Omega^0(\partial\Sigma,\mathfrak{g})$. Because $(\eta-\xi)|_\beta = 0$, $\eta^\# = (\eta-\xi)^\# + \xi^\#$ is in $\ker\big(\sigma-\Phi^*\varpi\big)$.

We now prove $\iota\big(\eta^\#\big)\Phi^*\varpi = 0$. For any $\alpha \in T_A\mathcal{M}(\Sigma)$, by Proposition 3.2 and Lemma 3.3,

$$\begin{aligned}
\iota(\alpha)\iota\big(\eta^\#\big)\Phi^*\varpi &= \sum_{j=0}^{r}\iota(\alpha|_{V_j})\iota\big(\eta|_{V_j}{}^\#\big)\varpi_j \\
&= -\sum_{j=0}^{r}\iota(\alpha|_{V_j})\mathrm{d}\oint(A|_{V_j},\eta|_{V_j}) - \frac{1}{2}\iota(\alpha)\,\mathrm{Hol}^*\big(\theta_{\mathbb{H}}+\overline{\theta}_{\mathbb{H}},\eta|_\beta\big) \\
&= -\sum_{j=0}^{r}\oint(\alpha|_{V_j},\eta|_{V_j}) - \iota(\alpha)\iota\big(\xi^\#\big)\omega \\
&= -\sum_{j=0}^{r}\sum_{\lambda_j=1}^{m_j}\int_{b_{\lambda_j}^j}^{b_{\lambda_{j+1}}^j}\big(\alpha|_{V_j},\mathrm{Ad}\big(\mathrm{Hol}_s^{b_{\lambda_j}^j}(A|_{V_j})^{-1}\big)\xi\big(b_{\lambda_j}^j\big)\big)\mathrm{d}s \\
&= \sum_{j=0}^{r}\sum_{\lambda_j=1}^{m_j}\int_{b_{\lambda_j}^j}^{b_{\lambda_{j+1}}^j}\big(\mathrm{Ad}\big(\mathrm{Hol}_s^{b_{\lambda_j}^j}(A)\big)\alpha|_{V_j},\xi\big(b_{\lambda_j}^j\big)\big) \\
&= \sum_{j=0}^{r}\sum_{\lambda_j=1}^{m_j}\big(\iota(\alpha|_{V_j})\big(\mathrm{Hol}_{b_{\lambda_{j+1}}^j}^{b_{\lambda_j}^j}\big)^*\overline{\theta},\xi\big(b_{\lambda_j}^j\big)\big) \\
&= \frac{1}{2}\sum_{j=0}^{r}\sum_{\lambda_j=0}^{m_j}\big(\iota(\alpha|_{V_j})\big(\mathrm{Hol}_{b_{\lambda_{j+1}}^j}^{b_{\lambda_j}^j}\big)^*\overline{\theta},\xi\big(b_{\lambda_j}^j\big)\big) \\
&\quad + \frac{1}{2}\sum_{j=0}^{r}\sum_{\lambda_j=0}^{m_j}\big(\iota(\alpha|_{V_j})\big(\mathrm{Hol}_{b_{\lambda_{j+1}}^j}^{b_{\lambda_j}^j}\big)^*\overline{\theta},\mathrm{Ad}\big(\mathrm{Hol}_{b_{\lambda_{j+1}}^j}^{b_{\lambda_j}^j}\big)\xi\big(b_{\lambda_{j+1}}^j\big)\big) \\
&= \iota(\alpha)\frac{1}{2}\,\mathrm{Hol}^*\big(\theta_{\mathbb{H}}+\overline{\theta}_{\mathbb{H}},\xi|_\beta\big) = 0.
\end{aligned}$$

Considering $\iota\big(\eta^\#\big)\Phi^*\varpi = 0$ and $\eta^\# \in \ker\big(\sigma-\Phi^*\varpi\big)$, we obtain $\iota\big(\eta^\#\big)\sigma = \iota\big(\eta^\#\big)\big((\sigma-\Phi^*\varpi\big)+\Phi^*\varpi\big) = 0$. Since σ is weak non-degenerate, $\eta^\#$ is zero. Therefore, we conclude ${\xi|_\beta}^\# = {(\xi-\eta)|_\beta}^\# + {\eta|_\beta}^\# = 0$. □

Remark 4.6.

(1) If the group Γ is trivial, we can choose any finite subset of $\partial\Sigma$ as the set of base points. Then the space $M(\Sigma)$ is a straightforward generalization of the space in [2, Theorem 9.1].

(2) Although Theorem 4.5 follows directly from Theorem 3.8, we include a proof because the almost same argument applies to the case where G is not simply connected.

4.6. **Fixed points of $M(X)$.** Taking fixed point objects of $(M(\Sigma),\omega,\mu)$, we obtain a twisted q-Hamiltonian $\big(G^\beta\big)^\Gamma$-space $\big(M(\Sigma)^\Gamma,\omega^\Gamma,\mu^\Gamma\big)$. We denote $\underline{\Sigma} := \Sigma/\Gamma$ and $\beta_{\underline{\Sigma}} := \pi(\beta)$, where $\pi\colon \Sigma \to \underline{\Sigma}$ is the covering map.

Taking a holonomy yields the following correspondence: $M(\Sigma) \simeq \mathrm{Hom}(\Pi_1(\Sigma,\beta),G)$. This correspondence is Γ-equivariant for an action on $\mathrm{Hom}(\Pi_1(\Sigma,\beta),G)$ $(\varphi\cdot\rho)(\gamma) = \varphi\rho(\gamma\cdot\varphi)$, $\varphi\in\Gamma$, $\rho\in\mathrm{Hom}(\Pi_1(\Sigma,\beta),G)$, $\gamma\in\Pi_1(\Sigma,\beta)$. Therefore, we obtain a bijection $M(\Sigma)^\Gamma \simeq \mathrm{Hom}(\Pi_1(\Sigma,\beta),G)^\Gamma$.

Now we want to consider the space $\mathrm{Hom}(\Pi_1(\Sigma,\beta),G)^\Gamma$ as the space on $\underline{\Sigma}$. Let $G_{\underline{\Sigma}}$ be a group bundle $(\Sigma\times G)/\Gamma$ on $\underline{\Sigma}$ with a fiber G. By fixing a complete system of representatives I of β for Γ-action, we obtain the monodromy representation of $G_{\underline{\Sigma}}$ denoted by $\mathrm{mon}_{\underline{\Sigma},I}\colon \Pi_1(\underline{\Sigma},\beta_{\underline{\Sigma}})\to\Gamma$ (cf. Appendix 5). We denote the space of representations twisted by the morphism $\mathrm{mon}_{\underline{\Sigma},I}$ by

$$\mathrm{Hom}_{\mathrm{mon}_{\underline{\Sigma},I}}(\Pi_1(\underline{\Sigma},\beta_{\underline{\Sigma}}),G) := \big\{\underline{\rho}\in\mathrm{Hom}(\Pi_1(\underline{\Sigma},\beta_{\underline{\Sigma}}),\Gamma\ltimes G) \mid \mathrm{pr}\circ\underline{\rho} = \mathrm{mon}_{\underline{\Sigma},I}\big\}.$$

Then, the covering map $\pi\colon\Sigma\to\underline{\Sigma}$ induces a bijection

$$\mathrm{Hom}_{\mathrm{mon}_{\underline{\Sigma},I}}(\Pi_1(\underline{\Sigma},\beta_{\underline{\Sigma}}),G) \simeq \mathrm{Hom}(\Pi_1(\Sigma,\beta),G)^\Gamma.$$

Therefore, we obtain a bijection $M(\Sigma)^\Gamma \simeq \mathrm{Hom}_{\mathrm{mon}_{\underline{\Sigma},I}}(\Pi_1(\underline{\Sigma},\beta_{\underline{\Sigma}}),G)$. By choosing a system of free generators of $\Pi_1(\underline{\Sigma},\beta_{\underline{\Sigma}})$, the space $\mathrm{Hom}_{\mathrm{mon}_{\underline{\Sigma},I}}(\Pi_1(\underline{\Sigma},\beta_{\underline{\Sigma}}),G)$ is diffeomorphic to a product of G-bitorsors. Since G is connected, this product of G-bitorsors is connected. Therefore, the tuple $\big(M(\Sigma)^\Gamma,\omega^\Gamma,\mu^\Gamma\big)$ is a twisted quasi-Hamiltonian $\big(G^\beta\big)^\Gamma$-space. Also $\mathrm{Hom}_{\mathrm{mon}_{\underline{\Sigma},I}}(\Pi_1(\underline{\Sigma},\beta_{\underline{\Sigma}}),G)$ carries a twisted quasi-Hamiltonian $G^{\beta_{\underline{\Sigma}}}$-structure. The above discussion is summarized in the following theorem.

Theorem 4.7. *The representation space* $\mathrm{Hom}(\Pi_1(\Sigma,\beta),G)$ *has a structure of a* Γ*-compatible quasi-Hamiltonian* G^β*-space. By taking the set of* Γ*-fixed points of* $\mathrm{Hom}(\Pi_1(\Sigma,\beta),G)$*, we obtain a structure of a twisted quasi-Hamiltonian* $G^{\beta_{\underline{\Sigma}}}$*-space on* $\mathrm{Hom}_{\mathrm{mon}_{\underline{\Sigma},I}}(\Pi_1(\underline{\Sigma},\beta_{\underline{\Sigma}}),G)$.

5. Twisted group bundles and monodromy

5.1. **Setting.** Let Σ be an oriented compact 2-manifold with boundary. Let Γ be a finite group. We consider a left Γ-action on G and a free right Γ-action on Σ. We define $\underline{\Sigma} := \Sigma/\Gamma$, trivial group bundle $G_\Sigma := \Sigma\times G$ on Σ, a group bundle $G_{\underline{\Sigma}} := (\Sigma\times G)/\Gamma$ on $\underline{\Sigma}$.

Let $\beta_{\underline{\Sigma}}$ be a finite subset of $\underline{\Sigma}$, and let β_Σ be a pre-image of $\beta_{\underline{\Sigma}}$ for the quotient map $q\colon\Sigma\to\underline{\Sigma}$. We denote the fundamental groupoid by $\Pi_\Sigma := \Pi_1(\Sigma,\beta_\Sigma)$ and $\Pi_{\underline{\Sigma}} := \Pi_1(\underline{\Sigma},\beta_{\underline{\Sigma}})$.

5.2. **The monodromy of $G_{\underline{\Sigma}}$.** The trivial group bundle G_Σ has the trivial flat Ehresmann connection $\mathcal{H}_\Sigma$. We denote its monodromy by mon_Σ. Since G_Σ is trivial as a group bundle, under the natural identification $G_b\simeq G$ for all $b\in\beta_\Sigma$, we obtain the trivial monodromy representation $\mathrm{mon}_{\Sigma,\beta}\colon\Pi_\Sigma\to\mathrm{Aut}(G), \gamma\mapsto 1$.

Since $\mathcal{H}_\Sigma$ is Γ-invariant under the diagonal action of Γ on G_Σ, $\mathcal{H}_\Sigma$ induces a flat Ehresmann connection $\mathcal{H}_{\underline{\Sigma}}$ on $\underline{\Sigma}$. We denote its monodromy by $\mathrm{mon}_{\underline{\Sigma}}$. For any $\gamma\in\Pi_{\underline{\Sigma}}$, $\mathrm{mon}_{\underline{\Sigma}}(\gamma)\in\mathrm{Isom}(G_{s(\gamma)},G_{t(\gamma)})$.

Fix a complete system of representatives $I\subset\beta_\Sigma$ for the Γ-action. We construct a Γ-valued representation of $\mathrm{mon}_{\underline{\Sigma}}$. For any $\gamma\in\Pi_{\underline{\Sigma}}$, $i(s(\gamma))\in I$ denotes a representative of $q^{-1}(s(\gamma))$, and $i(t(\gamma))\in I$ denotes a representative of $q^{-1}(t(\gamma))$. Let $\tilde{\gamma}$ be a lift of $\gamma\in\Pi_{\underline{\Sigma}}$ starting at $i(s(\gamma))$, then there exists a unique element $\varphi_I(\gamma)\in\Gamma$ such that $t(\tilde{\gamma}) = i(t(\gamma))\cdot\varphi_I(\gamma)$. Set $\mathrm{mon}_{\underline{\Sigma},I}(\gamma) := \varphi_I(\gamma)$, we obtain the monodromy representation $\mathrm{mon}_{\underline{\Sigma},I}\colon\Pi_{\underline{\Sigma}}\to\Gamma$.

5.3. **The G-valued representation of $\Pi_{\underline{\Sigma}}$.** We denote

$$\mathrm{Hom}_{\mathrm{mon}_{\underline{\Sigma},I}}(\Pi_{\underline{\Sigma}},G) := \{\underline{\rho}\in\mathrm{Hom}(\Pi_{\underline{\Sigma}},\Gamma\ltimes G) \mid \mathrm{pr}_\Gamma\circ\underline{\rho} = \mathrm{mon}_{\underline{\Sigma},I}\}.$$

In this appendix, we construct a bijection

$$\mathrm{Hom}(\Pi_\Sigma, G)^\Gamma \to \mathrm{Hom}_{\mathrm{mon}_{\underline{\Sigma},I}}(\Pi_{\underline{\Sigma}}, G), \qquad \rho \mapsto \underline{\rho}_I.$$

A Γ-action on $\mathrm{Hom}(\Pi_\Sigma, G)$ is defined by $(\varphi \cdot \rho)(\tilde{\gamma}) = \varphi\rho(\tilde{\gamma} \cdot \varphi)$, $\rho \in \mathrm{Hom}(\Pi_\Sigma, G)$, $\varphi \in \Gamma$, $\tilde{\gamma} \in \Pi_\Sigma$. For any $\rho \in \mathrm{Hom}(\Pi_\Sigma, G)^\Gamma$ and $\gamma \in \Pi_{\underline{\Sigma}}$, we denote the lifting of γ starting at $i(s(\gamma))$ by $\tilde{\gamma}$. We define $\underline{\rho}_I(\gamma) := (\mathrm{mon}_{\underline{\Sigma},I}(\gamma), \rho(\tilde{\gamma}))$.

Proposition 5.1. *$\underline{\rho}_I \colon \Pi_{\underline{\Sigma}} \to \Gamma \ltimes G$ is a groupoid homomorphism.*

Proof. For any $\gamma_1, \gamma_2 \in \Pi_{\underline{\Sigma}}$ satisfying $s(\gamma_1) = t(\gamma_2)$, $\widetilde{(\gamma_1\gamma_2)} = (\tilde{\gamma_1} \cdot \mathrm{mon}_{Y,I}(\gamma_2))\tilde{\gamma_2}$. Therefore,

$$\underline{\rho}_I(\gamma_1\gamma_2) = (\mathrm{mon}_{Y,I}(\gamma_1\gamma_2), \rho(\widetilde{\gamma_1\gamma_2})) = (\mathrm{mon}_{Y,I}(\gamma_1)\,\mathrm{mon}_{Y,I}(\gamma_2), \rho(\tilde{\gamma_1} \cdot \mathrm{mon}_{Y,I}(\gamma_2))\rho(\tilde{\gamma_2}))$$

□

$$= (\mathrm{mon}_{Y,I}(\gamma_1)\,\mathrm{mon}_{Y,I}(\gamma_2), (\rho(\tilde{\gamma_1}) \cdot \mathrm{mon}_{Y,I}(\gamma_2))\rho(\tilde{\gamma_2})) = \underline{\rho}_I(\gamma_1)\underline{\rho}_I(\gamma_2).$$

It is easy to see that the above correspondence is bijective.

6. A C^∞ structure of fixed point sets

In this appendix, we show that each connected component of the fixed point set N^Γ for a C^∞-action of a finite group Γ on a Banach manifold N is a Banach submanifold of N.

Fix a point $x \in N^\Gamma$, a Γ-action on the tangent space T_xN is induced by the Γ-action on N. We denote the averaging by $\mathrm{ave}_x \colon T_xN \to (T_xN)^\Gamma$, $X \mapsto \frac{1}{|\Gamma|}\sum_{\varphi\in\Gamma} \varphi \cdot X$. Because the Γ-action on T_xN is top-linear, the averaging is also top-linear.

Lemma 6.1. $\ker(\mathrm{ave}_x)$ *and* $(T_xN)^\Gamma$ *are closed subspaces of* T_xN. *Furthermore, a sequence*

$$0 \to \ker(\mathrm{ave}_x) \hookrightarrow T_xN \twoheadrightarrow (T_xN)^\Gamma \to 0$$

is split.

Proof. Since $(T_xN)^\Gamma$ is an intersection of closed subspaces $(T_xN)^\Gamma = \bigcap_{\varphi\in\Gamma} \ker(\varphi - 1)$, the space $(T_xN)^\Gamma$ is closed. Because ave_x is top-linear, the kernel $\ker(\mathrm{ave}_x)$ is a closed subspace. For any $X \in T_xN$,

$$(\mathrm{ave}_x)^2(X) = \mathrm{ave}_x\left(\frac{1}{|\Gamma|}\sum_{\varphi\in\Gamma} \varphi \cdot X\right) = \frac{1}{|\Gamma|}\sum_{\varphi\in\Gamma} \mathrm{ave}_x(\varphi \cdot X) = \frac{1}{|\Gamma|}\sum_{\varphi\in\Gamma} \varphi \cdot X = \mathrm{ave}_x(X).$$

Therefore, we have $T_xN = \ker(\mathrm{ave}_x) \oplus (T_xN)^\Gamma$. □

Suppose that N^Γ is connected. Fix an element $x \in N^\Gamma$. We show that there exists a good chart around $x \in N$. By Bochner's linearization theorem for compact Lie group actions on Banach manifolds [7, Proposition 3.1.4], there exists a Γ-invariant open neighborhood U of $x \in N^\Gamma$ and a Γ-equivariant diffeomorphism $\chi \colon U \xrightarrow{\sim} V \subset T_xN$ which satisfies $\mathrm{d}_x\chi = 1_{T_xN}$. Since $T_xN = \ker(\mathrm{ave}_x) \oplus (T_xN)^\Gamma$, the above neighborhood U can be written as $U = \chi^{-1}\big(V' \times V^\Gamma\big)$ where V' (resp. V^Γ) is a sufficiently small open neighborhood of 0 in $\ker(\mathrm{ave}_x)$ (resp. $(T_xN)^\Gamma$). Because χ is Γ-equivariant, we have $N^\Gamma \cap U \simeq V^\Gamma = V \cap (T_xN)^\Gamma$. Therefore, each connected component of N^Γ is a Banach submanifold of N.

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